

CE1: Savitzky-Golay Smoothing

- **Library Used:** `scipy.signal.savgol_filter`
- **Role:** Initial denoising while preserving local patterns and peaks.
- **Parameters Used:**
 - `window_length = 11`
 - `polyorder = 3`

Rationale:

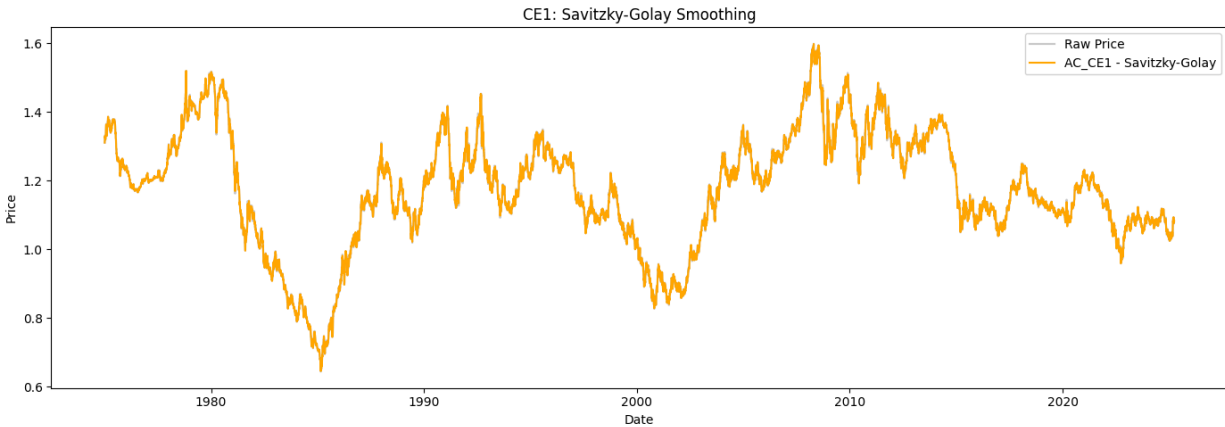
Savitzky-Golay fits a polynomial to local windows of data using least squares. It's ideal for preserving high-frequency financial structures (spikes, dips) while removing short-term noise.

Behavior:

Produces a smooth but *shape-preserving* output. Acts as a cleaner version of the raw data—setting the stage for further smoothing.

Parameter Effects:

- **Increasing `window_length`** → More smoothing, risk of flattening features.
- **Decreasing `window_length`** → More noise retained, less stable output.
- **Increasing `polyorder`** → More flexible fit, but risk of overfitting.
- **Decreasing `polyorder`** → Oversimplifies shape, especially around turning points.



✅ CE2: Gaussian Kernel Smoothing (Nadaraya-Watson)

- **Library Used:** Custom kernel smoother (via NumPy)
- **Role:** Smooths CE1 output using a Gaussian-weighted average.
- **Parameters Used:**
 - `bandwidth = 3.0`

📌 Rationale:

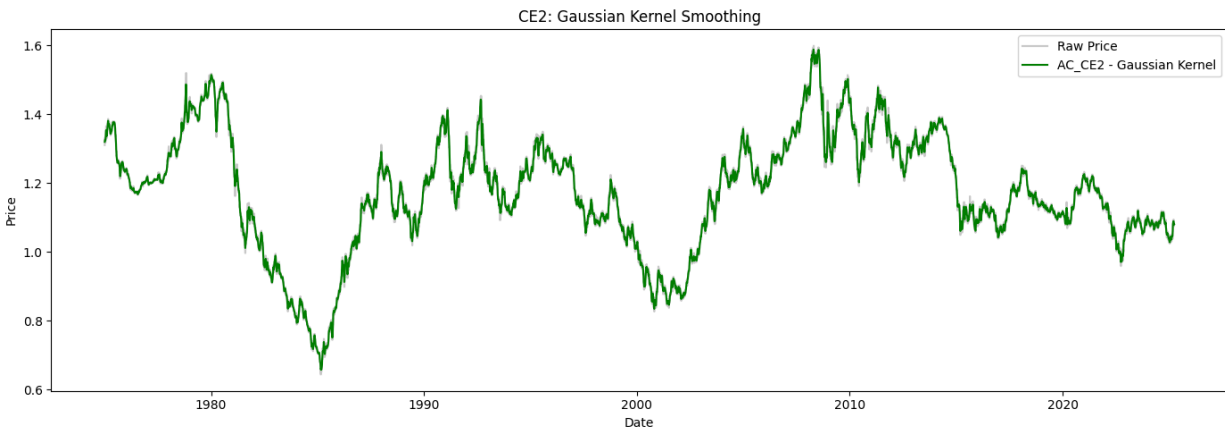
This method computes the smoothed value as a weighted average of nearby points, where weights decay based on distance using a Gaussian function.

📈 Behavior:

Strips away remaining high-frequency volatility from CE1, leading to a visually cleaner trend curve with minimal lag.

🔧 Parameter Effects:

- Increasing **bandwidth** → More smoothing, loss of local detail.
- Decreasing **bandwidth** → Reacts more to noise, retains volatility.



✓ CE3: Kernel-Weighted Polynomial Regression

- **Library Used:** Custom implementation (Gaussian weights + poly regression)
- **Role:** Adds nonlinear structure awareness, allowing dynamic curvature.
- **Parameters Used:**
 - **degree = 2**
 - **bandwidth = 8**

📌 Rationale:

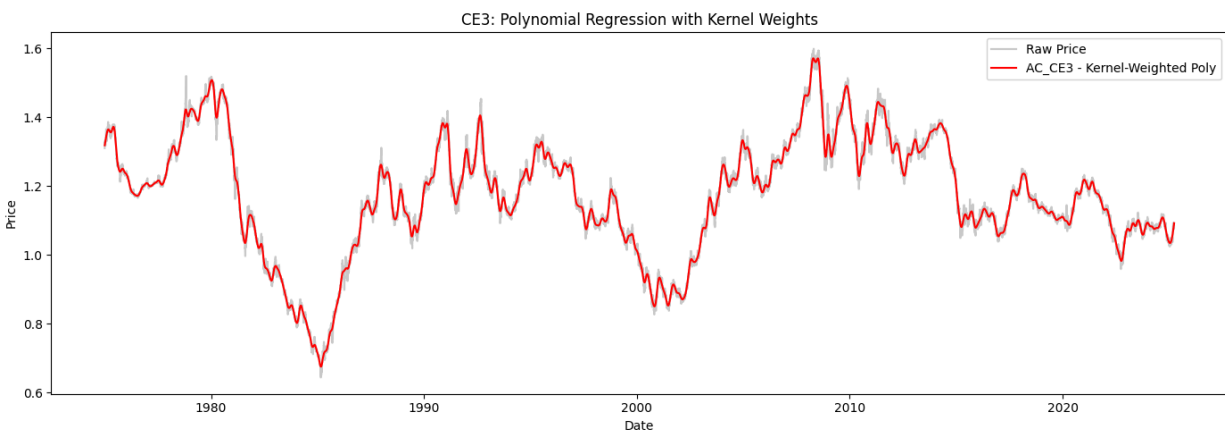
A polynomial (quadratic) is locally fitted with weights centered at each point. This lets the curve respond to curvature more intelligently than global polynomials.

📈 Behavior:

Generates local curvatures (bowls, domes) where appropriate. Unlike isotonic or moving averages, this captures trend momentum and reversal shapes.

Parameter Effects:

- Increasing **degree** → Allows more complexity, but risks oscillation.
- Decreasing **degree** → Too rigid, might miss shape detail.
- Increasing **bandwidth** → More generalized smoothing across wider neighborhoods.
- Decreasing **bandwidth** → Higher sensitivity to local structure, potentially more noise.



CE4: Gaussian Kernel Regression (Final Pass)

- **Library Used:** Custom kernel regression (Gaussian smoothing)
- **Role:** Final polishing layer to remove any jaggedness or edge noise.
- **Parameters Used:**
 - **bandwidth** = 6

Rationale:

A second pass of kernel regression stabilizes CE3 while maintaining the nuanced movements gained from the previous step. This is where the final Alternative Curve (AC) is output.

Behavior:

Gives a fluid, natural-looking curve closely aligned to mid-term market behavior. Handles dips and rallies without jitter or constraint.

Parameter Effects:

- **Higher bandwidth** → Final output becomes smoother but flatter.
- **Lower bandwidth** → Curve becomes overly reactive to short-term fluctuations, potentially introducing noise again.

